

Test Case 1: Single Producer and Single Consumer

```
[grant@HongOS] - [2026-04-03 03:29:28 PM] - [~/src/repos/graknow/eecs-university/eecs-3540/hw5] 10
[0] ./buffer 3 1 1
producer produced 933203417
consumer consumed 933203417
```

Test Case 2: Multiple Producers and Multiple Consumers

```
[grant@HongOS] - [2026-04-03 03:29:33 PM] -
[0] ./buffer 5 2 2
producer produced 1192184773
producer produced 550444591
consumer consumed 1192184773
consumer consumed 550444591
producer produced 89399630
producer produced 1427714729
consumer consumed 89399630
consumer consumed 1427714729
```

Test Case 3: Buffer Full Condition

```
[grant@HongOS] - [2026-04-03 03:40:03 PM] -
[0] ./buffer 5 3 1
producer produced 694181192
producer produced 2044888687
producer produced 266924103
consumer consumed 694181192
producer produced 383313099
consumer consumed 2044888687
producer produced 1929697126
consumer consumed 266924103
producer produced 1554669920
producer produced 154495058
```

Test Case 4: Buffer Empty Condition

```
[grant@HongOS] - [2026-04-03 03:40:15 PM] -
[0] ./buffer 5 1 3
producer produced 1716451127
consumer consumed 1716451127
producer produced 1257018889
consumer consumed 1257018889
```

Test Case 5: Rapid Alternation Between Producers and Consumers

```
[0] ./buffer 5 5 5
producer produced 408460152
producer produced 957376169
consumer consumed 408460152
producer produced 36410553
producer produced 287563279
producer produced 1345206022
consumer consumed 957376169
consumer consumed 36410553
producer produced 934648175
consumer consumed 287563279
producer produced 1663529870
consumer consumed 1345206022
producer produced 32551251
consumer consumed 1663529870
consumer consumed 957376169
producer produced 1094009864
producer produced 2076171901
consumer consumed 32551251
producer produced 1082467080
consumer consumed 1094009864
consumer consumed 2076171901
```